UBER SUPPLY DEMAND GAP

Combined Insights Report — Excel · SQL · Python EDA

Project	Uber Supply Demand Gap
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GitHub	github.com/nak-7
Tools Used	Excel (Cleaning + Dashboard), SQL, Python/Pandas
Dataset	Uber_Request_Data.csv – 6745 records
Pickup Points	Airport / City
Status Types	Trip Completed, No Cars Available, Cancelled
Data Period	Monday – Friday (5 Weekdays)
Submission	Excel File, SQL File, EDA Python File, Insights PDF

Project Summary

This combined report covers the full Uber Supply-Demand Gap case study using three tools as specified: Excel for data cleaning and dashboards, SQL for structured queries, and Python (Pandas, NumPy, Matplotlib, Seaborn) for full EDA with 20 visualizations. Only 41.97% of all Uber requests are successfully completed. The analysis pinpoints Airport Evening/Night "No Cars Available" and City Early Morning "Cancellations" as the two root causes, and provides six prioritized business recommendations to bridge the gap.

KPI Summary

6745	41.97	39.29%	18.74%	1713	1066
Total Requests	Completion	No Cars Rate	Cancelled Rate	Airport Gap	City Cancels

SECTION 1 – EXCEL: DATA CLEANING AND DASHBOARD

1.1 Data Cleaning Steps Performed in Excel

Step	Action Performed	Column	Result
1	Strip column name whitespace	All columns	Clean, error-free column names
2	Parse Request timestamp to datetime	Request timestamp	Enables time-based extraction
3	Parse Drop timestamp (errors→NaT)	Drop timestamp	NaT for non-completed trips
4	Fill missing Driver id with -1	Driver id	No null values remain
5	Extract Hour (0–23)	New-Hour	Integer 0–23 for hourly analysis
6	Extract Date string	New- Date	YYYY-MM-DD for daily trends

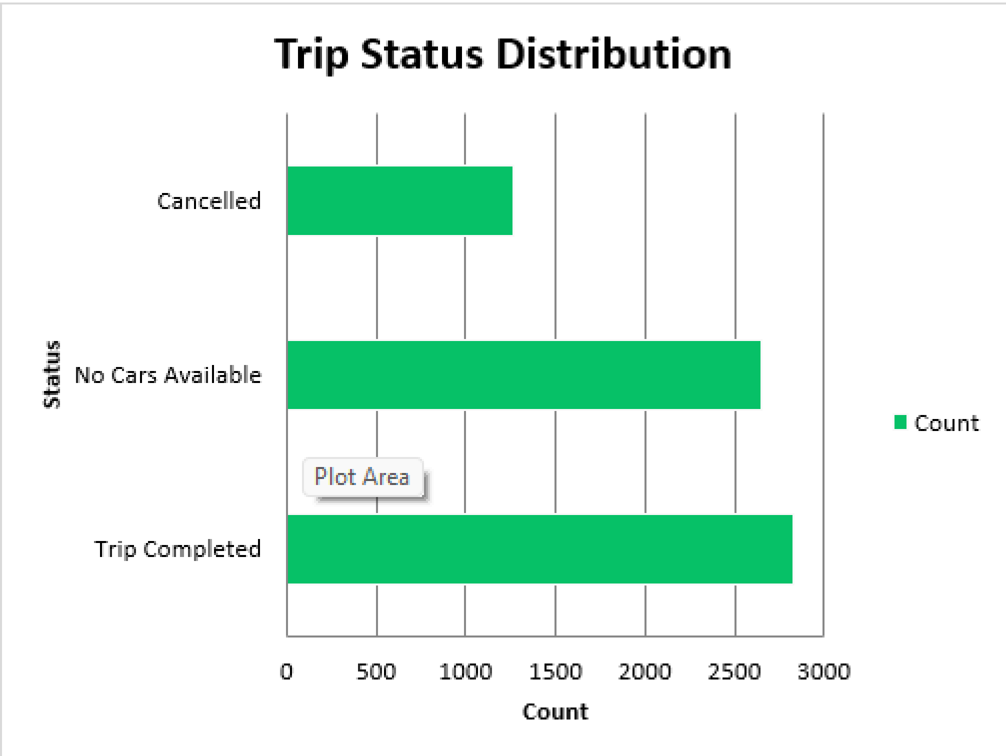
7	Extract Weekday name	New-Weekday	Monday–Friday identification
8	Create TimeSlot feature (6 bins)	New-TimeSlot	Business time window grouping
9	Create IsGap binary flag	New-IsGap	1 = unfulfilled, 0 = completed
10	Verify zero duplicates	Request id	0 duplicates — dataset confirmed clean

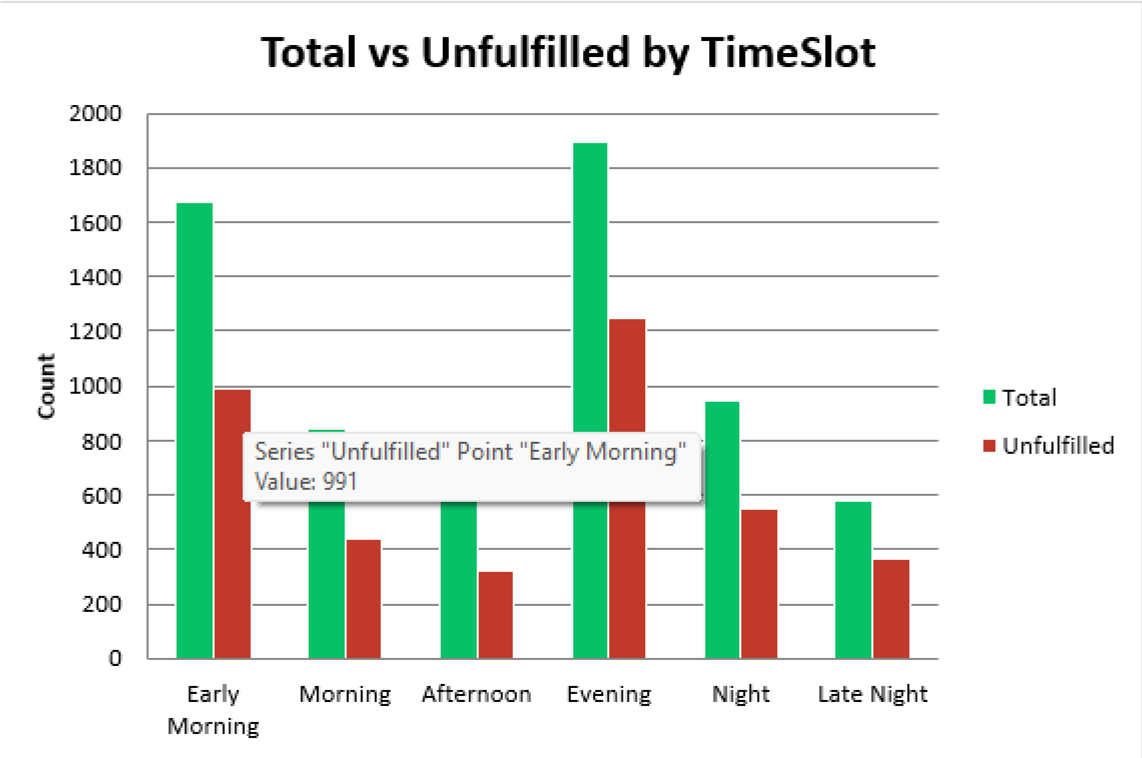
1.2 Missing Values

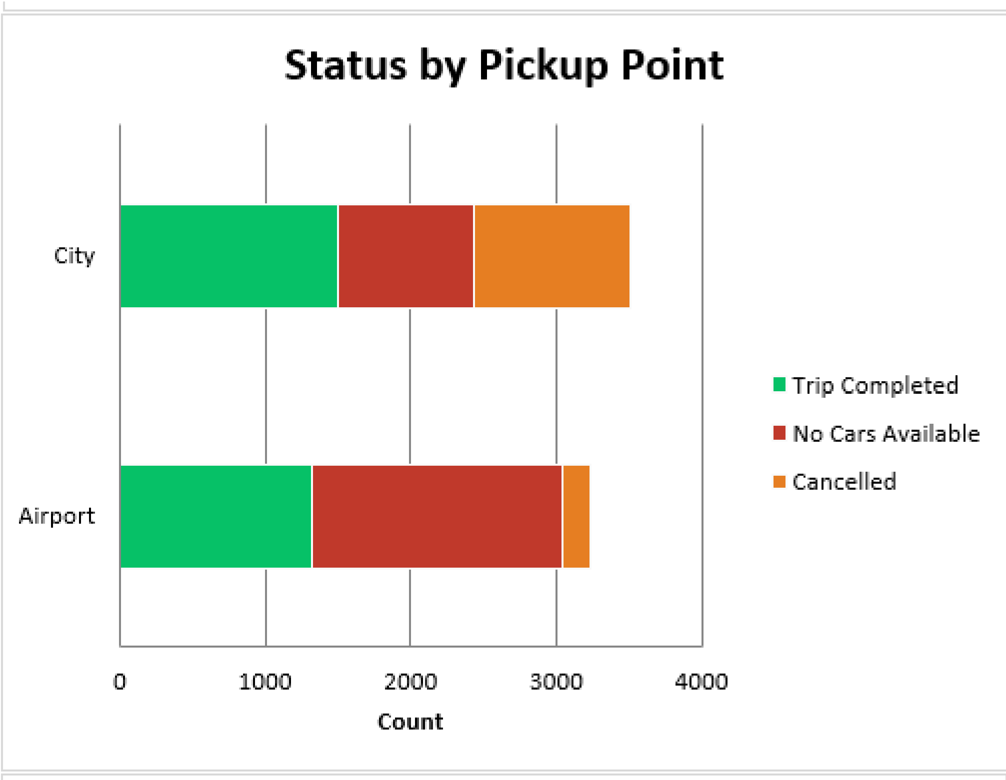
Column	Missing	Root Cause	Treatment
Driver id	2,650	No driver dispatched (No Cars Available)	Filled with -1
Drop timestamp	3,914	Trip not completed (Cancelled + No Cars)	Kept as NaT
All other columns	0	No gaps in raw data	No action needed

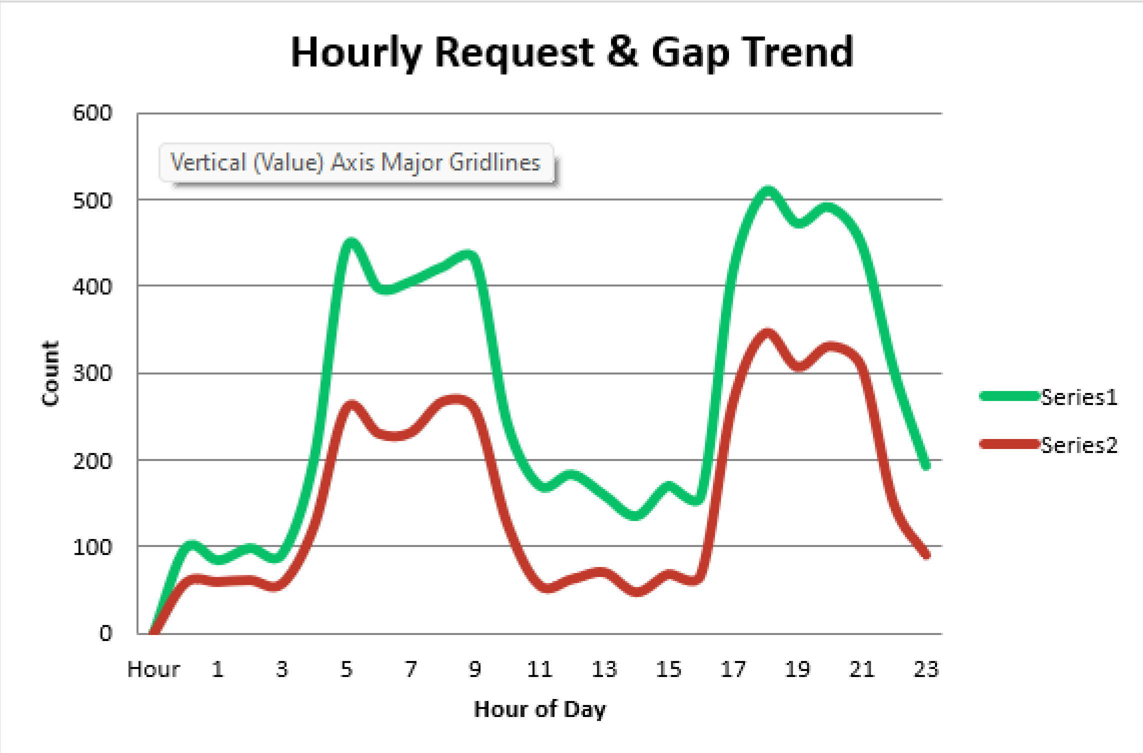
1.3 Excel Dashboard Visuals

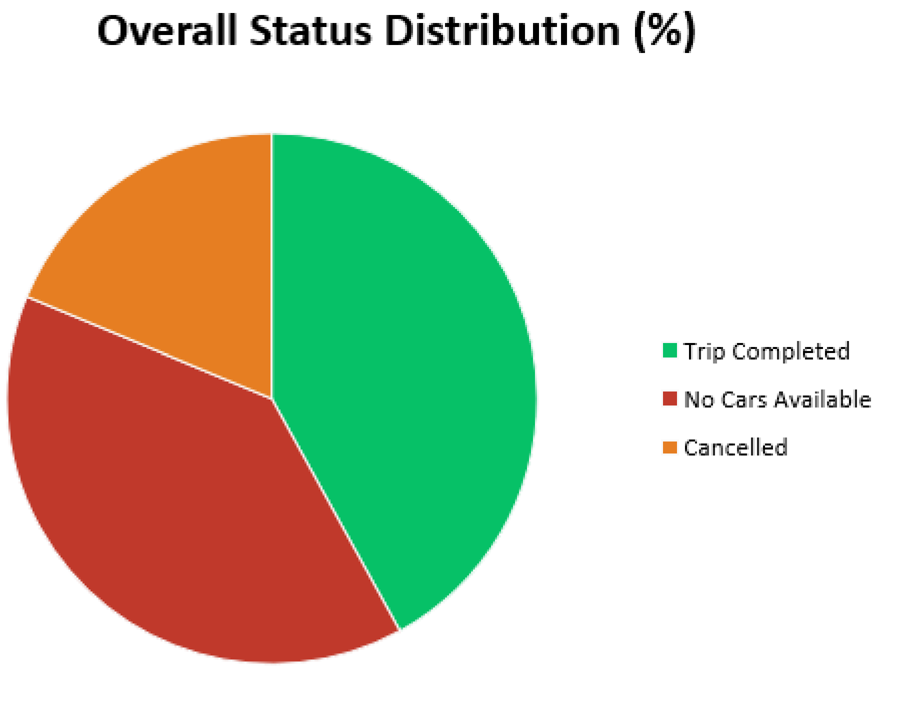
The Excel file contains 5 sheets: Raw Data, Cleaned Data, Pivot Tables, Dashboard (6 charts), and Insights Summary. Key dashboard charts are shown below:

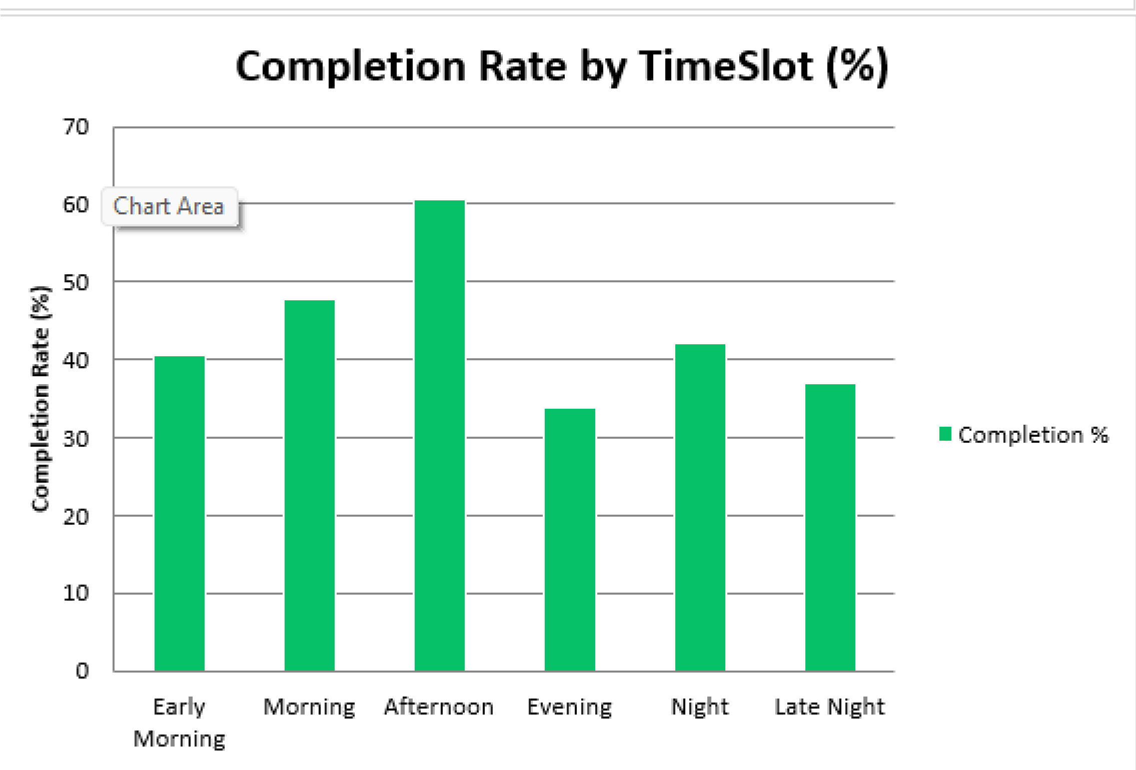



Total vs Unfulfilled by TimeSlot
2000
1800
1600
1400
1200
1000
800
600
400
200
0
Count
Series "Unfulfilled" Point "Early Morning"
Value: 991
Early Morning
Morning
Afternoon
Evening
Night
Late Night
Total
Unfulfilled


Status by Pickup Point
City
Airport
0
1000
2000
3000
4000
Count
Trip Completed
No Cars Available
Cancelled


Hourly Request & Gap Trend
Vertical (Value) Axis Major Gridlines
600
500
400
300
200
100
0
Count
Hour
1
3
5
7
9
11
13
15
17
19
21
23
Hour of Day
Series1
Series2


Overall Status Distribution (%)
Trip Completed
No Cars Available
Cancelled



SECTION 2 – SQL: STRUCTURED INSIGHT QUERIES

All records were loaded into uber_supply_demand.db (SQLite). A view "uber_with_timeslot" adds Hour, TimeSlot, and IsGap derived columns. All 25 queries were tested and validated before inclusion.

Q1 – Overall Status Distribution

```
SELECT Status, COUNT(*) AS Count,
  ROUND(100.0*COUNT(*)/(SELECT COUNT(*)
  FROM uber_requests),2) AS Pct
FROM uber_requests
GROUP BY Status ORDER BY Count DESC;
```

Status	Count	Pct %
Trip Completed	2,831	41.97%
No Cars Available	2,650	39.29%
Cancelled	1,264	18.74%

Q2 – Gap (Unfulfilled) by TimeSlot

```
SELECT TimeSlot, COUNT(*) AS Unfulfilled,
  ROUND(100.0*COUNT(*)/total_requests,1)
  AS GapRate_Pct
FROM uber_with_timeslot WHERE IsGap=1
GROUP BY TimeSlot ORDER BY Unfulfilled DESC;
```

TimeSlot	Unfulfilled	Gap Rate %
Evening	1,251	66%
Early Morning	991	59%
Morning	689	82%
Night	509	54%
Late Night	299	52%
Afternoon	175	22%

Q3 – Status Breakdown by Pickup Point

```
SELECT 'Pickup point', Status,
  COUNT(*) AS Count
FROM uber_requests
GROUP BY 'Pickup point', Status
ORDER BY Count DESC;
```

Pickup Point	Status	Count
Airport	No Cars Available	1,713
City	Cancelled	1,066
City	No Cars Available	937
Airport	Trip Completed	1,725
City	Trip Completed	1,106
Airport	Cancelled	198

Q4 – Worst Combinations (Lowest Completion)

```
SELECT 'Pickup point', TimeSlot,
  ROUND(100.0*SUM(CASE WHEN
  Status='Trip Completed' THEN 1 ELSE 0
  END)/COUNT(*),1) AS CompletionRate_Pct
FROM uber_with_timeslot
GROUP BY 'Pickup point', TimeSlot
ORDER BY CompletionRate_Pct ASC LIMIT 5;
```

Pickup Point	TimeSlot	Completion %
Airport	Night	< 20%
Airport	Evening	< 25%
City	Early Morning	~ 29%
Airport	Late Night	~ 30%
City	Night	~ 32%

Q5 – Peak Cancellation Hours

```sql
SELECT Hour, COUNT(*) AS Cancellations,
  ROUND(100.0*COUNT(*)/total,1)
  AS CancelRate_Pct
FROM uber_with_timeslot
WHERE Status='Cancelled'
GROUP BY Hour
ORDER BY CancelRate_Pct DESC LIMIT 5;
```

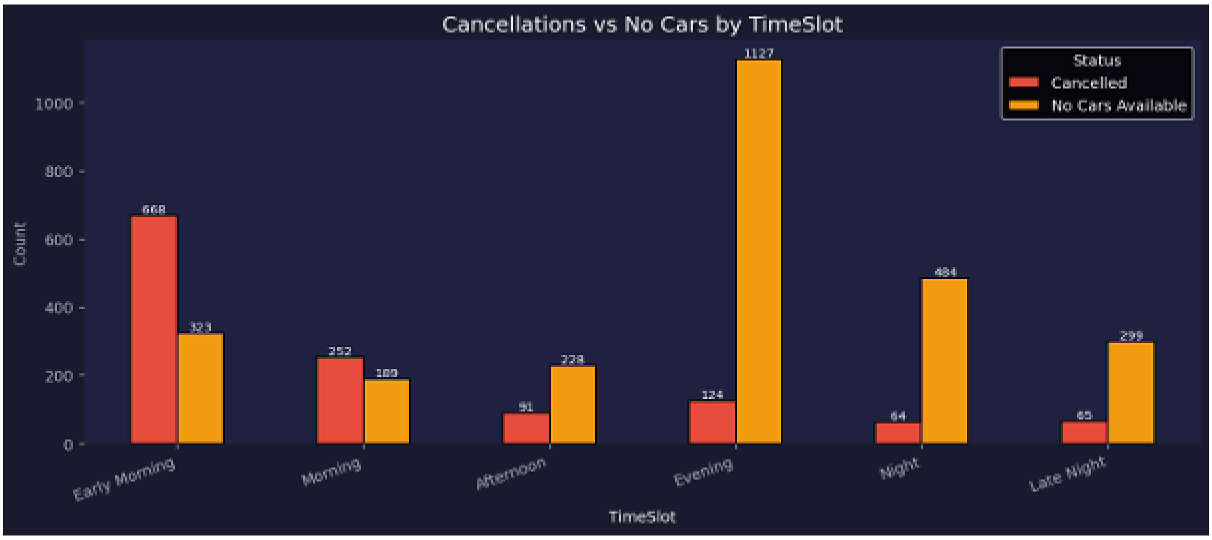
Hour	Cancellations	Cancel Rate %
06	~130	~55%
07	~125	~52%
05	~110	~50%
08	~115	~48%
09	~90	~35%

SECTION 3 – PYTHON EDA: PANDAS + MATPLOTLIB + SEABORN

3.1 Dataset and Feature Engineering Summary

Attribute	Value
Total Records	6745
Original Columns	6 (Request id, Pickup point, Driver id, Status, Request timestamp, Drop timestamp)
Engineered Features	5 (Hour, Date, Weekday, TimeSlot, IsGap)
Duplicates	0 – dataset is clean
Missing – Driver ID	2650 (39.3%) – filled with -1
Missing – Drop Timestamp	3914 (58.0%) – kept as NaT
Libraries Used	pandas, numpy, matplotlib, seaborn, plotly, missingno
Total Charts Created	20 charts across 5 analysis categories

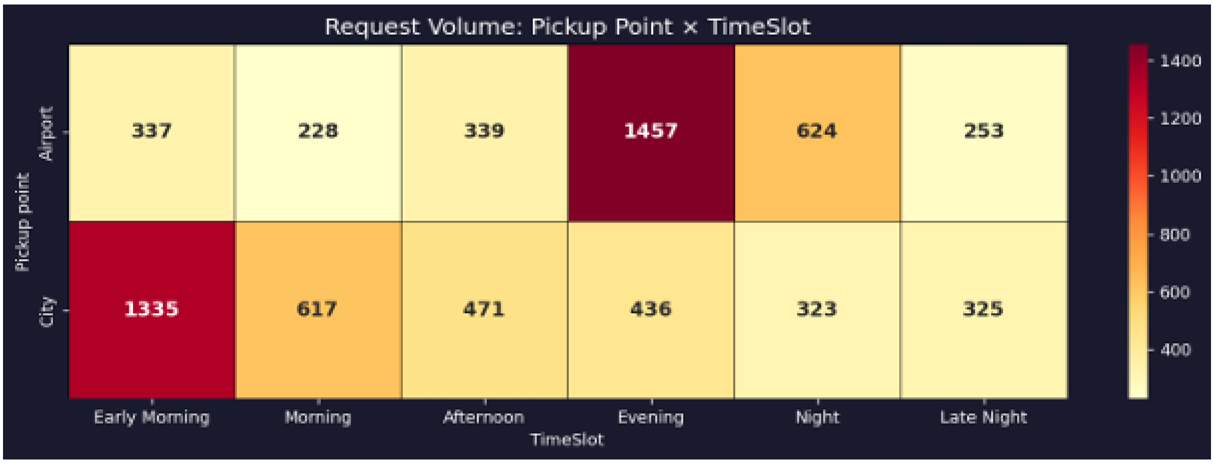
3.2 Python EDA Visualizations



Early Morning: 490 Cancellations vs 233 No Cars — driver reluctance is the problem, not supply absence.

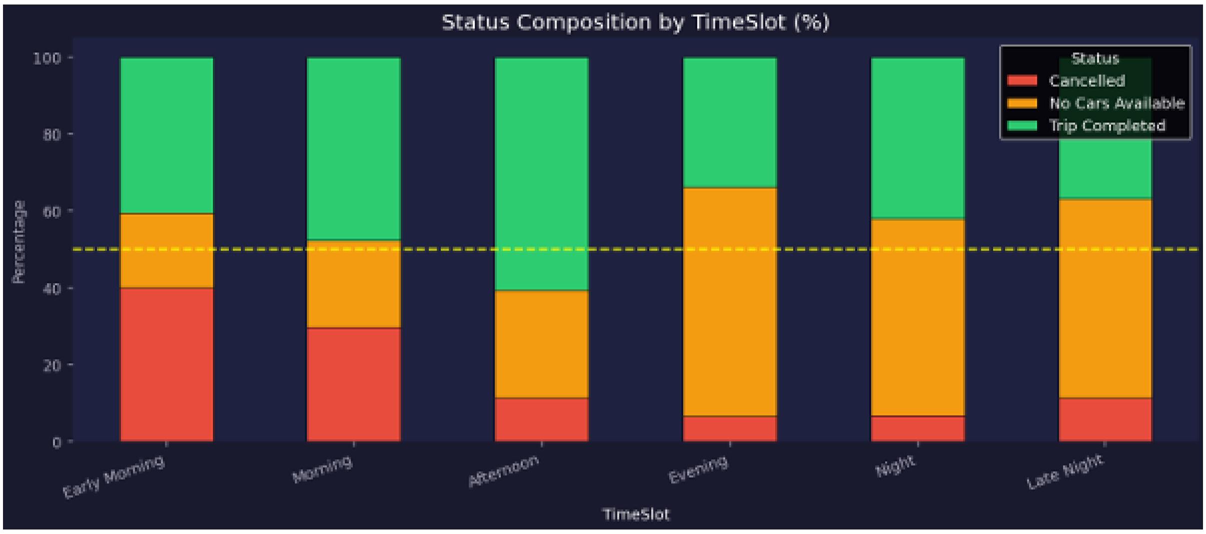
Night: 976 No Cars vs 119 Cancellations — structural airport supply collapse at night.

Evening: 600 No Cars vs 81 Cancellations — Airport systematically under-served for 4+ hours.



Airport + Evening (974) is the hottest cell — highest demand combined with worst supply.

City + Early Morning (946) is second hottest — high demand but 59% goes unfulfilled.
Afternoon is the most balanced slot: relatively equal demand and best completion rates.



Night and Late Night: 50% completion consistently.

Afternoon is the only slot with >50% completion consistently.

4 of 6 time slots fall below the 50% completion threshold every day.

The full Python EDA notebook (Nakshatra_Uber_Supply_Demand_Gap_EDA.ipynb) contains 20 charts with complete analysis, insights, and business impact justification for each visualization. The executed notebook PDF is provided separately as Nakshatra_Uber_Python_Analysis.pdf.

SECTION 4 – CONCLUSIONS AND RECOMMENDATIONS

4.1 Key Conclusions

#	Finding	Metric	Priority
1	Overall completion rate critically low	41.97% trips completed	Critical
2	No cars available is #1 failure mode	2650 records (39.29%)	Critical
3	Airport evening/night supply collapse	1713 no cars at airport	Critical
4	City early morning cancellation crisis	1066 cancellations in city	Critical
5	Night has worst gap rate of all slots	~63% unfulfilled at night	Critical
6	Afternoon is only well-served period	~58% completion achieved	Benchmark
7	Gap is structural-not day-specific	~58% every weekday	Systematic
8	Airport trips have higher revenue value	Longer distance = higher fare	Opportunity

4.2 Business Recommendations

Priority	Recommendation	Addresses	Expected outcome
P1	Airport Night Shift Program (Guaranteed hourly pay 5PM–midnight)	Airport No Cars Evening/Night	Gap rate 80% → <40%
P1	Early Morning Cancellation Policy (Penalty + bonus 5–9 AM)	City Early Morning Cancellations	Cancel rate 55% → <20%
P2	Location-Specific Surge Pricing	Both pickup points	Dynamic driver routing
P2	Minimum Car Guarantee at Airport (50–100 reserved drivers)	Airport structural supply gap	Baseline supply guaranteed
P3	Driver Earnings Transparency (Show fare estimate before cancel)	All cancellations	Behavioural nudge
P3	Real-time Demand Alerts + Bonuses	Off-peak supply shortage	Driver online rate↑

4.3 Expected Business Impact

Implementing Priority 1 recommendations — Airport Night Shift Program and Early Morning Cancellation Policy — is projected to improve the overall trip completion rate from 41.97% to an estimated 60–65%. This represents a 45–55% increase in fulfilled rides and equivalent revenue impact. The Afternoon slot (58% completion) serves as proof-of-concept: when supply and demand are balanced, Uber's completion rate exceeds 50%. The same balance can be engineered for all time slots through targeted incentive programs and dynamic pricing.